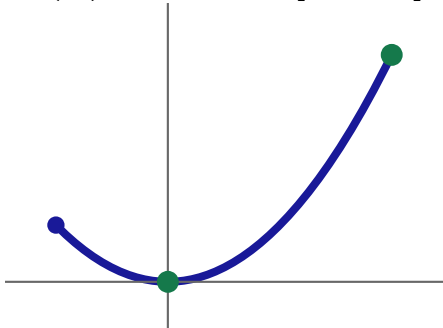
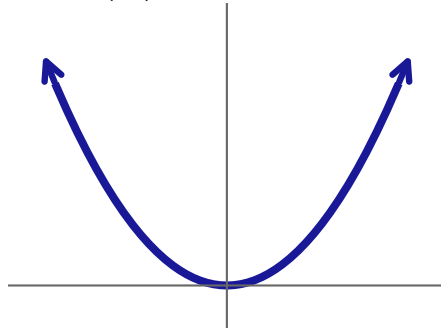


$$f(x) = x^2 \text{ on } [-1, 2]$$



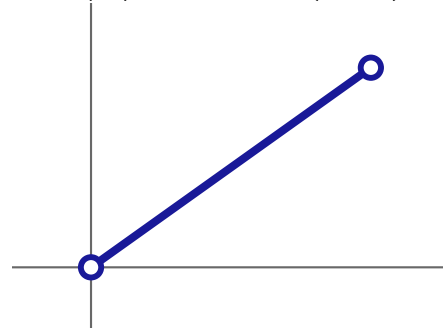
D compact, f continuous
 $\implies$ max and min attained

$$f(x) = x^2 \text{ on } \mathbb{R}$$



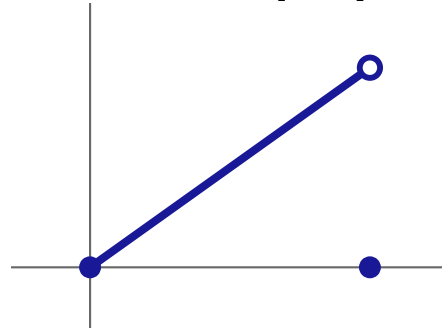
D not bounded
 $\implies$ no maximum

$$f(x) = x \text{ on } (0, 1)$$



D not closed
 $\implies$ neither attained

$$\text{jump on } [0, 1]$$



f not continuous
 $\implies$ no maximum